Vulnerability Patching via Structure-Aware Retrieval-Augmented Generation

Master's Thesis Progress

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What Is a Vulnerability?

NIST FIPS 200 [1]

A **vulnerability** is a weakness in an information system, system security procedures, internal controls, or implementation that could be exploited by a threat source.

Example:
Use-After-Free

The program references memory after it has been freed, leading to undefined behavior.

```
struct sock *alloc_socket(
    struct socket *sock) {
  struct sock *sk;

  sk = sk_alloc();        // allocate

  sock->sk = sk;          // alias created

  if (!chan_create()) {
    sk_free(sk);          // free original
    return NULL;     // BUG: alias dangling!
  }
  return sk;
}
```

Inspired by CVE-2024-56605 (Linux kernel) — a real-world use-after-free vulnerability.

Tracing the Vulnerability (1/4): Allocation

```
struct sock *alloc_socket(
    struct socket *sock) {
  struct sock *sk;

  sk = sk_alloc();          // allocate

  sock->sk = sk;            // alias created

  if (!chan_create()) {
    sk_free(sk);            // free original
    return NULL;
  }
  return sk;
}
```

- **`sk` is allocated** — a new object is created on the heap

Tracing the Vulnerability (2/4): Alias Creation

```
struct sock *alloc_socket(
    struct socket *sock) {
  struct sock *sk;

  sk = sk_alloc();          // allocate

  sock->sk = sk;            // alias created

  if (!chan_create()) {
    sk_free(sk);            // free original
    return NULL;
  }
  return sk;
}
```

- `sk` is allocated
- **`sock->sk` becomes an alias** — two pointers now reference the same object

Tracing the Vulnerability (3/4): Free

```
struct sock *alloc_socket(
    struct socket *sock) {
  struct sock *sk;

  sk = sk_alloc();          // allocate

  sock->sk = sk;            // alias created

  if (!chan_create()) {
    sk_free(sk);            // free original
    return NULL;
  }
  return sk;
}
```

- `sk` is allocated
- `sock->sk` becomes an alias
- **On error, `sk` is freed** — the object is deallocated

Tracing the Vulnerability (4/4): Dangling Pointer

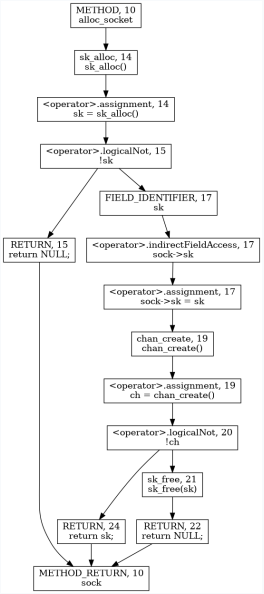
```
struct sock *alloc_socket(
    struct socket *sock) {
  struct sock *sk;

  sk = sk_alloc();          // allocate

  sock->sk = sk;            // alias created

  if (!chan_create()) {
    sk_free(sk);            // free original
    return NULL;      // BUG: alias dangling!
  }
  return sk;
}
```

- sk is allocated
- sock->sk becomes an alias
- On error, sk is freed
- **sock->sk still points to freed memory!**

Pattern:
Free without nullifying all aliases
⇒ **Use-After-Free**

This pattern leaves a structural fingerprint in the code graph.

Code Property Graph: Full Joern [2] Output

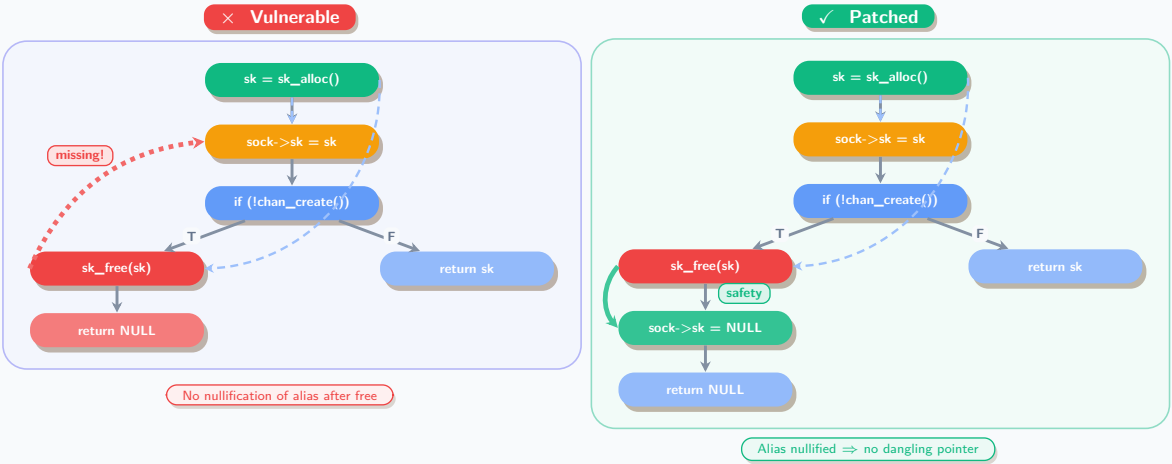


What the CPG encodes:

- **AST** — syntax structure
- **CFG** — execution order
- **DDG** — data flow (def → use)
- **CDG** — control dependencies

Full graph is detailed — let's simplify to the vulnerability-relevant subgraph.

Structure Reveals the Vulnerability



Takeaway: The structural difference between vulnerable and patched code is a single edge/node — vulnerabilities leave fingerprints in the graph.

Motivation: Why Not Just Train a Graph Classifier?

Prior work exploits the same insight — code structure reveals vulnerabilities:

- Train GNNs / graph kernels on labeled CPGs or PDGs
- Learn to classify subgraphs as vulnerable vs. safe
- Devign [3], ReVeal [4], IVDetect [5], ...

But these approaches hit a wall:

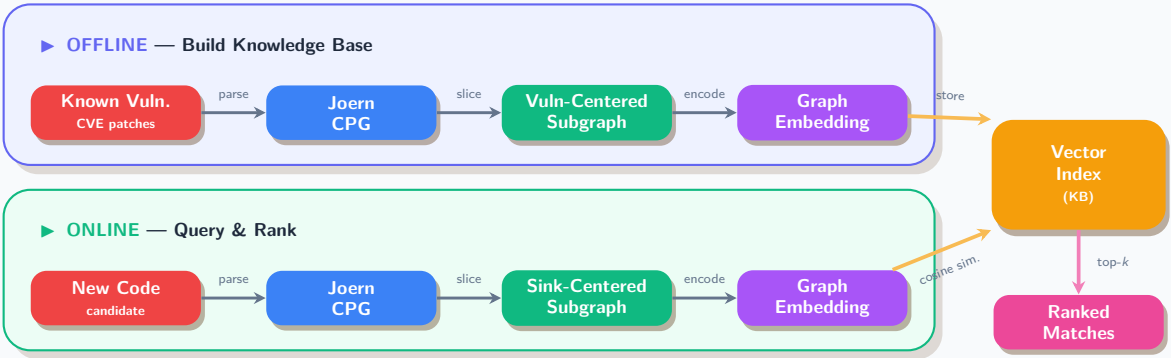
- Require large labeled datasets (expensive to curate)
- Require retraining to *unseen* vulnerability types
- Treat each vulnerability independently — no knowledge reuse

Our alternative: Instead of *classifying*, **retrieve**.
Store known vulnerability patterns as graph embeddings in a knowledge base, then rank new code by structural similarity — no retraining needed.

Research Questions

RQ1: How effective are graph-based embeddings at retrieving relevant vulnerability fix patterns from the knowledge base?
RQ2: What are the trade-offs of using Graph-RAG in the vulnerability patching pipeline?

System Architecture: Graph-RAG Pipeline



Key idea: Identical pipeline for both tracks — only the *input* differs. **Offline** indexes *known* vulnerability patterns; **Online** queries *unseen* code against them.

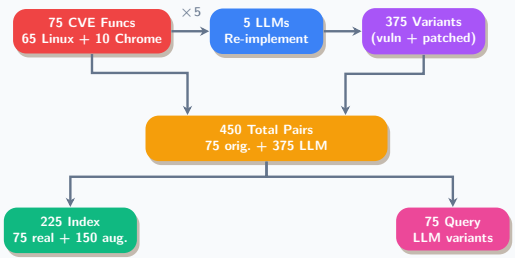
Feasibility Study: AutoPatch Dataset (n=225)

Goal: Validate the concept using a controlled dataset and define baselines.

AutoPatch Dataset:

- 75 CVE functions (65 Linux Kernel, 10 Chromium)
- 5 LLMs re-implement each $\rightarrow$ 375 variant pairs

Can structural graph embeddings capture vulnerability patterns as effectively as pretrained CodeBERT?



Evaluation Metrics

Hit@1

Success rate at rank 1

$$\frac{1}{|Q|}\sum_{q} \mathbb{1}[\text{rank}_q = 1]$$

Fraction of queries where the correct match is ranked **first**.

Hit@5

Success rate in top 5

$$\frac{1}{|Q|}\sum_{q} \mathbb{1}[\text{rank}_q \leq 5]$$

Fraction of queries where the correct match appears in **top 5**.

MRR

Mean Reciprocal Rank

$$\frac{1}{|Q|}\sum_{q} \frac{1}{\text{rank}_q}$$

Average inverse rank across queries — rewards **higher** placements.

CWE Recall

Type coverage

$$\frac{|\text{CWE}_{\text{retrieved}}|}{|\text{CWE}_{\text{index}}|}$$

Fraction of distinct CWE types correctly retrieved **at least once**.

Embedding Pipeline: From Graph to Vector

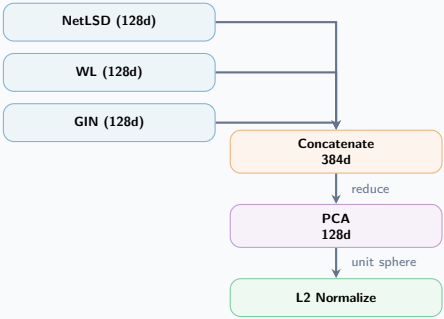
Structure-Only Embedders (no training required):

Embedder	Dim	Mechanism
NetLSD [6]	128	Heat kernel trace on graph Laplacian spectrum (100 timescales 10^{-2}–10^{2})
WL [7]	128	Weisfeiler-Lehman color refinement (4 iters) → sum pool → linear proj.
GIN [8]	128	3-layer Graph Isomorphism Network (random weights, frozen) → mean+sum readout

Semantic Baseline:

Embedder	Dim	Mechanism
CodeBERT [9]	768	Pretrained transformer; CLS token on linearized code

Combined Pipeline:



PCA: fit on index set, retains >95% variance.
All final vectors: **128d**, L2-normalized.

Node features: 11-class CPG node type (one-hot) — METHOD, CALL, IDENTIFIER, LITERAL, RETURN, CONTROL_STRUCTURE, ...

Phase 1: Feasibility — AutoPatch Dataset (n=225)

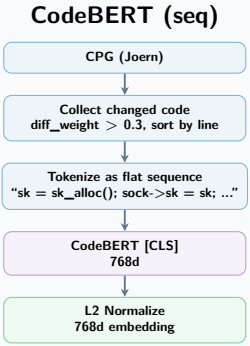
Goal: Validate that structural embeddings match CodeBERT performance.

Embedder	H@1	H@5	MRR	CWE R.
Structure-Only				
Combined	**0.8904**	**0.9589**	**0.9162**	**0.4978**
GIN	0.7260	0.8767	0.7879	0.4427
WL	0.6986	0.7945	0.7363	0.4386
CodeBERT pretrained				
CodeBERT (baseline)	0.1533	0.2200	0.1826	0.2933
CodeBERT (seq)	0.7534	0.8630	0.7927	0.4121
CodeBERT (pat)	0.7123	0.8356	0.7677	0.4147

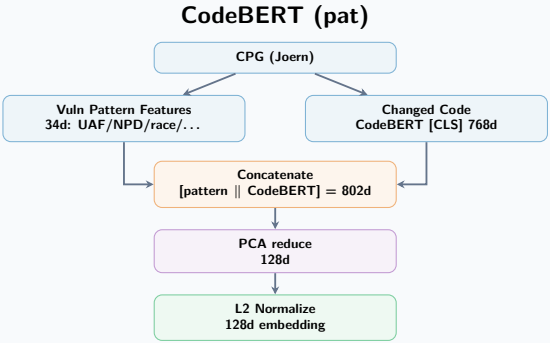
Source: `experiments/output/checkpoint5/20260505_141347_experiment_f59ab1/results.json`

RQ1: validated — structural *graph* representations are *effective* for vulnerability pattern retrieval in a synthetic dataset.

CodeBERT Embeddings — Sequence vs. Pattern



Text-only: structure is **discarded**.



Fuses **graph structure** with semantics.

Phase 2: Generalization — CVEfixes Dataset

Question: Does the approach generalize to real-world vulnerabilities?

Real-World Dataset

CVEFixes [10] datasets is large dataset with real-world CVEs collected from the U.S. National Vulnerability Database, patches, and code changes.

Metric	Value
CVEs	11,873
Fix commits	12,107
File changes	51,342
Method changes	277,948
Repositories	4,249

Sampling criteria

Structurally distinshable CWEs (e.g. UAF vs. OOB) to test pattern matching, not just code similarity

Balanced superclass(CWE) up to 20 entries and subclasses (CVE) to ensure retrieval feasibility

Parameter	Value
Total pairs	136
Index set	109
Query set	27
Unique CVEs (index)	60
Unique CVEs (query)	25
CWE classes	7

Phase 2: Pipeline Verification Results

Same-CWE Retrieval (n=109 index, 27 query)

Embedder	Hit@1	Hit@5	MRR	CWE Recall
GIN	0.259	0.778	0.315	0.202
Combined	0.111	0.778	0.223	0.189
CodeBERT (baseline)	TODO?			
CodeBERT (seq)	**0.556**	**0.926**	**0.555**	**0.264**
CodeBERT (pat)	**0.519**	**0.889**	**0.549**	**0.276**

Source: `cvefixes_experiments/output/pipeline_verification/results.json`

RQ1 validated on real-world dataset with same-CWE retrieval up to 93% at k=5.

Per-CWE Breakdown (CodeBERT Pattern, CWE Hit@k)

CWE	N	@1	@5	@10
Integer Overflow (190)	4	75%	100%	100%
Input Validation (20)	4	75%	75%	100%
Resource Consump. (400)	3	67%	100%	100%
Use After Free (416)	4	50%	100%	100%
NULL Ptr Deref (476)	5	40%	80%	80%
Race Condition (362)	3	33%	100%	100%
OOB Write (787)	4	25%	75%	75%
Overall	**27**	**51.9%**	**88.9%**	**92.6%**

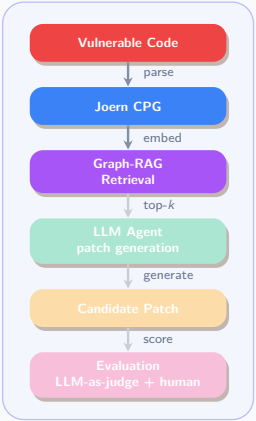
Source: same file (CodeBERT Pattern cell)

Highlights areas for improvement in embedding design or slicing strategy.

Next Steps: Research Question 2

RQ2: What are the trade-offs of using Graph-RAG in the vulnerability patching pipeline?

1. Integrate the retrieval layer with patch agent
2. Run comparative experiments against baseline approaches:
   - zero-shot open-source models
   - AutoPatch
3. Validate the end-to-end patching pipeline with industry professionals.



Questions?

Lívia Cavalcanti | graph-rag/

Appendix: CPG Deep Dive

The Code Property Graph encodes four types of relationships:

- **AST:** Abstract syntax tree (structure of code)
- **CFG:** Control flow graph (which statements execute in sequence)
- **DDG:** Data dependency graph (variable definitions and uses)
- **CDG:** Control dependency graph (which statements depend on conditionals)

Joern exposes all four via traversals; KB retrieval compares the combined structure to find similar vulnerabilities.

Appendix: Joern Query - Finding the Pattern

Goal: Find `free()` calls with dangling aliases.

```
val uafCandidates = cpg.call.name(".*free.*")
  .map { freeCall =>
    val freedVar = freeCall.argument(1).code
    val aliases = freeCall.method.assignment
      .where(_.source.isIdentifier.name(freedVar))
      .target.code.l
    val freeLineNo = freeCall.lineNumber.get
    val nullified = freeCall.method.assignment
      .where(_.lineNumber.map(_ > freeLineNo))
      .target.code.l
    val dangling = aliases.filterNot(nullified.contains)
    (freeCall, dangling)
  }
```

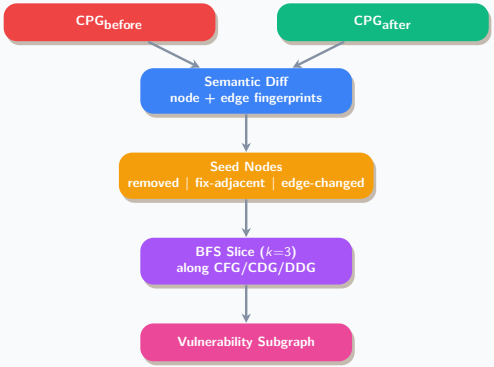
Output: `[UAF] sk_free(sk) - dangling:  sock->sk`

Appendix: Vulnerability-Aware Code Slicing

Goal: Extract a minimal subgraph capturing the vulnerability context from the full CPG.

Algorithm (3 phases):

1. **Semantic Node Diff** — Match nodes by fingerprint (`labelV, CODE, LINE_NUMBER`) between pre-/post-patch CPGs. Identify removed, added, and fix-adjacent nodes.
2. **Semantic Edge Diff** — Detect structural changes (rewired control/data flow).
3. **Bounded Program Slice** — BFS k=3 hops along flow edges (CFG, CDG, REACHING_DEF, PDG, DDG) from seed nodes.



Diff Label	Weight
removed	1.0
fix_adjacent	0.8
edge_changed	0.6
context	0.2

Appendix: CodeBERT Pattern Embedding — Detail

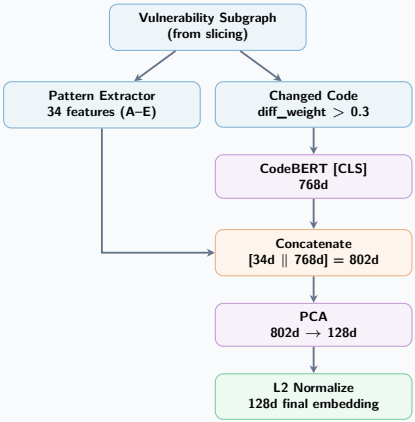
34d Feature Vector — 5 Groups:

All features require CPG edge connectivity (REACHING_DEF, CDG, CFG).

Grp	Dim	Features & Mechanism
A: Vuln Flow (8)	A1	UAF: free $\xrightarrow{\text{REACH\_DEF}}$ identifier (2-hop)
	A2	NPD: ptr deref w/o null-check in CDG preds
	A3	Unchecked return: CALL $\not\rightarrow$ CONTROL_STRUCT
	A4	Lock/unlock imbalance: $\lvert L-U\rvert/(L+U+1)$
	A5	Unguarded arithmetic: no bounds in CDG neighbourhood
	A6	Use w/o def: IDENTIFIER w/o incoming REACH_DEF
	A7	Alloc/free imbalance: $\lvert A-F\rvert/(A+F+1)$
	A8	Cast in data flow: cast $\xleftrightarrow{\text{REACH\_DEF}}$ changed
B (8)	B1–8	Edge type distribution among changed nodes AST, CFG, CDG, REACH_DEF, REF, ARG, RECV, CALL
C (6)	C1–6	Boundary flow: changed→ctx vs ctx→changed per CFG, CDG, REACHING_DEF (2 dirs × 3 types)
D (6)	D1–6	Diff topology: changed frac, density, #components, mean/max degree, internal vs crossing edge ratio
E (6)	E1–6	Node role distribution of changed nodes CALL, CTRL_STRUCT, ID, LITERAL, BLOCK, OTHER

Node classification uses regex on `CODE` attr (e.g. `free|kfree|vfree` for free nodes). Flow features traverse CPG edges.

Full Pipeline:



Appendix: Joern Generated graph

TODO: add graphml example

References I

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[2] Joern Project. *Joern – The Bug Hunter's Workbench*. 2024. URL: https://joern.io.

[3] Yaqin Zhou et al. "Devign: Effective Vulnerability Identification by Learning Comprehensive Program Semantics via Graph Neural Networks". In: *Advances in Neural Information Processing Systems (NeurIPS)*. 2019.

[4] Saikat Chakraborty et al. "Deep Learning based Vulnerability Detection: Are We There Yet?" In: *IEEE Transactions on Software Engineering* (2022).

[5] Yi Li, Shaohua Wang, and Tien N. Nguyen. "Vulnerability Detection with Fine-Grained Interpretations". In: *Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*. 2021.

References II

[6] Anton Tsitsulin et al. “NetLSD: Hearing the Shape of a Graph”. In: *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. KDD ’18. London, United Kingdom: Association for Computing Machinery, 2018, pp. 2347–2356. ISBN: 9781450355520. DOI: 10.1145/3219819.3219991. URL: https://doi.org/10.1145/3219819.3219991.

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[9] Zhangyin Feng et al. “CodeBERT: A Pre-Trained Model for Programming and Natural Languages”. In: *Findings of the Association for Computational Linguistics: EMNLP 2020*. 2020.

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